

Action-Conditioned Observability-Risk Modulated PPO for LiDAR-Reduced UAV Pursuit

Anonymous Authors

Abstract—High-speed unmanned aerial vehicle pursuit is commonly studied with privileged relative states or simplified sensing assumptions, but onboard pursuit must make decisions from limited, noisy, and intermittently sparse sensor evidence. This paper studies LiDAR/range-image reduced-information pursuit, where the pursuer cannot use target ground truth, semantic masks, or simulator-only target identifiers at policy execution time. We formulate short-horizon target observability loss as a sensing-continuity risk, learn action-conditioned risk estimates from privileged rollouts, and use the resulting signal to modulate proximal policy optimization rather than applying a post-hoc action filter. The resulting framework is designed to separate the contribution of reduced sensing, state-only risk prediction, heuristic risk shaping, and action-conditioned observability-risk modulation through a staged baseline and ablation protocol. The paper defines the problem setting, literature position, method interface, and evaluation protocol while reserving numerical performance claims for completed experiments.

Index Terms—UAV pursuit, partial observability, LiDAR, observability risk, proximal policy optimization, safe reinforcement learning.

I. INTRODUCTION

UNMANNED aerial vehicle (UAV) pursuit-evasion is a core problem for autonomous airspace security, cooperative robotics, and adversarial motion planning. Recent reinforcement-learning and planning systems have improved UAV pursuit, interception, and multi-agent maneuvering in high-speed or unknown environments [5], [8], [22], [31], [32], [34]. However, many pursuit policies are still evaluated under privileged or heavily processed observations. In real onboard pursuit, the pursuer must decide not only how to reduce range, but also how to keep the target detectable through a limited sensor aperture during aggressive motion.

Limited visual field, visibility-based pursuit-evasion, bearing-only pursuit, target tracking, and vision-based capture studies show that target loss and reacquisition are first-order components of robotic pursuit [2], [10], [17], [18], [28], [35], [38], [39]. These works make clear that finite sensing cannot be treated as an implementation detail. A high-speed pursuer can choose an action that is locally good for closing distance while simultaneously pushing the target toward the edge of the detectable region, causing future observations to collapse. This motivates an explicit model of how candidate actions affect near-term target observability.

The proposed direction is also related to constrained and safe reinforcement learning, including constrained policy optimization, Lagrangian methods, recovery policies, shields, safety benchmarks, risk-sensitive policy gradients, and action-conditioned risk gating [1], [3], [9], [13], [20], [25], [27]. These methods establish that risk-aware policy learning and

action gating are not new in the general sense. The distinction pursued here is narrower: the constrained variable is not collision or generic safety violation, but target observability loss in a LiDAR/range-image reduced-information UAV pursuit task.

This paper studies an action-conditioned observability-risk modulated PPO framework. A full-state pursuit policy provides an upper reference and privileged rollouts provide labels for short-horizon target loss. The online policy, however, receives only ego state, previous action, and LiDAR/range-image features. A learned risk model estimates both state risk and action-conditioned risk, denoted by $p_{\text{lost}}(s^{\text{red}})$ and $R_{\text{obs}}(s^{\text{red}}, u)$, where s^{red} is the reduced observation. The main policy remains a PPO controller that directly outputs collective-thrust and body-rate commands, while the learned risk signal is introduced inside policy learning rather than after the action has already been sampled.

The paper is organized around three contributions:

- a sensor-honest LiDAR/range-image reduced-information UAV pursuit formulation that excludes target truth, simulator semantic labels, and truth-derived bearing/range inputs at policy execution time;
- a short-horizon observability-loss risk model trained from privileged rollouts, including an action-conditioned form $R_{\text{obs}}(s^{\text{red}}, u)$ for ranking candidate actions by future target-loss risk;
- a staged PPO evaluation protocol that separates full-state pursuit, reduced-information pursuit, heuristic risk, state-only learned risk, and action-conditioned observability-risk modulation.

II. RELATED WORK

A. Finite-Sensing Pursuit and Target Reacquisition

Limited-field pursuit has been studied through multi-UAV visual-field constraints, visibility-based pursuit-evasion, and occlusion-aware control formulations [18], [28], [39]. These works directly motivate the target-loss and reacquisition part of our problem. The difference is that we do not model visibility only as a phase, reward, or control-barrier constraint. Instead, we learn a calibrated near-term observability-loss risk under reduced LiDAR/range-image evidence and use it as a structured input to policy learning.

Bearing-only pursuit, vision-based MAV capture, drone-chasing, and UAV target-tracking systems further show how difficult long-range or small-target pursuit becomes under onboard sensing [2], [10], [17], [35], [38]. These systems are important evidence that the sensing interface matters. Our setting differs by focusing on high-speed pursuit control under sparse range-image returns rather than on detector design, bounding-box refinement, or cooperative capture hardware.

B. UAV Pursuit-Evasion and Aerial Interception

Recent UAV pursuit-evasion studies include online planning in unknown environments, target-prediction-enhanced multi-UAV pursuit, high-speed maneuvering, agile quadrotor controllers, and reinforcement-learning-based aerial interception [5], [8], [22], [31], [34], [37]. These works define a strong domain baseline: they address pursuit performance, planning, deployment realism, or high-speed maneuvering. Our focus is different. We do not claim a new multi-UAV online planner or a new agile flight controller; we isolate how reduced sensing and action-conditioned observability risk affect pursuit policy learning.

Multi-agent and hierarchical pursuit-evasion methods improve cooperation, assignment, communication, scalability, and adversarial robustness [12], [14], [26], [29], [30], [36]. They are essential related work, but the present paper does not treat multi-agent cooperation as the central contribution. The planned comparison uses these papers to delimit scope and to prevent overclaiming novelty in pursuit-evasion reinforcement learning itself [32].

C. Constrained and Risk-Sensitive Reinforcement Learning

Constrained policy optimization, PID Lagrangian methods, Safety Gym, conservative adaptive penalties, distributional risk-sensitive actor-critic methods, and predictive CVaR policy gradients provide the general constrained and risk-sensitive RL background [1], [9], [15], [16], [20], [25]. Recovery RL and shielding further show that learned recovery zones or action shields can intervene when a task policy approaches unsafe states [3], [27]. Our method should therefore be compared against PPO-Lagrangian and shield/filter variants using the same observability-risk signal.

Risk-aware pursuit and high-speed safety-shielded flight are closer to the robotic application domain [24], [33]. Action-conditioned risk gating under partial observability is a particularly close method-level neighbor [13]. We therefore avoid claiming general action-conditioned risk gating as new. The proposed contribution is to change the risk semantics to target observability loss, learn that risk from privileged pursuit rollouts, and inject it into PPO under sensor-honest LiDAR/range-image inputs.

D. PPO and Privileged Training under Partial Observability

PPO is used as the base policy optimizer because it is a reliable on-policy method for continuous-control policy learning [23]. Asymmetric actor-critic and asymmetric DQN methods show how privileged simulator information can improve learning while the deployed actor receives partial observations [4], [19]. Recent informed asymmetric actor-critic work further studies which privileged signals are useful under partial observability [7]. In this paper, privileged information is used to label and supervise observability risk, not to provide target truth to the online actor.

Reduced-measurement interception, aerial-robot interception, and UAV autopilot studies demonstrate that EAAI-style contributions can combine reinforcement learning with flight-control engineering when the application assumption

is explicit [8], [11], [21]. Imperfect-information multi-UAV reasoning further supports the relevance of partial information in aerial autonomy [6]. The present paper follows this application-driven style: reinforcement learning is the training machinery, while the scientific claim is the action-conditioned observability-risk formulation for reduced-sensing pursuit.

III. METHOD

A. Problem Setting

We consider a pursuer UAV that tracks and captures a target UAV in simulation. The full-state baseline observes privileged geometric variables, including target-relative state. The reduced-information policy observes only an ego-state component, the previous collective-thrust/body-rate action, and a stack of LiDAR/range-image features. We denote this reduced observation by s_t^{red} . The policy outputs a normalized control command

$$u_t = [c_t, p_t, q_t, r_t] \in [-1, 1]^4, \quad (1)$$

where c_t is collective thrust and p_t, q_t, r_t are body-rate commands.

Let D_t denote whether the target is inside the pursuer's virtual detectable region at time t . A short-horizon observability-loss label is generated from privileged rollout metadata:

$$y_t = \mathbb{I} [\exists \tau \in [t, t + H] \text{ s.t. } D_{\tau:\tau+K_{\text{persist}}-1} = 0]. \quad (2)$$

The current experimental plan sets $H = 150$ and $K_{\text{persist}} = 10$, with horizon sensitivity planned for $H \in \{100, 150, 200\}$.

B. Observability-Risk Modulated PPO

The risk model estimates a state-only loss probability $p_{\text{lost}}(s_t^{\text{red}})$ and an action-conditioned score $R_{\text{obs}}(s_t^{\text{red}}, u)$. The latter is intended to compare candidate actions at the same reduced observation. The PPO policy remains the executed controller; the environment should receive the same sampled action whose log probability enters the PPO ratio. Risk is therefore used as an internal learning signal for actor features, action-distribution parameters, critic/risk values, or advantage shaping, rather than as a post-hoc replacement of the sampled action.

The main comparison ladder is: B0 full-state PPO as an upper reference, B1 reduced PPO without risk, B2 reduced PPO with a LiDAR-honest heuristic risk, B3 reduced PPO with learned state risk, and B4 reduced PPO with learned action-conditioned observability-risk modulation. Internal B4 ablations include risk concatenation, shuffled-action risk, PPO-Lagrangian with the same risk as a cost, shield/filter variants, privileged critic, and separate actor- and advantage-side risk modulation.

IV. EXPERIMENTAL DESIGN

The evaluation protocol separates offline risk quality from online pursuit performance. Offline risk evaluation will report discrimination and calibration metrics such as AUROC, AUPRC, Brier score, expected calibration error, reliability diagrams, and matched-state action-risk ranking accuracy. Online

evaluation will report success rate, time to capture, final target detectability, target-lost time, longest invisible streak, recovery count, recovery rate, control smoothness, action saturation, and visible-strike performance.

Because the method relies on sparse LiDAR/range-image evidence, experiments will also report target return count, range-image occupancy, dropout statistics, and sensitivity to noise, latency, angular resolution, reflectivity, and target range. The current plan treats 100 m, 150 m, and 200 m as sparse-return stress-test regimes rather than as a real-world long-range deployment guarantee.

V. CONCLUSION

This study frames high-speed UAV pursuit under reduced LiDAR/range-image sensing as an observability-risk-sensitive control problem. The intended contribution is not a generic safe PPO algorithm or a new pursuit-evasion benchmark, but a sensor-honest formulation in which privileged rollouts supervise action-conditioned target-loss risk and that risk modulates PPO learning. The staged B0–B4 protocol is designed to test whether action-conditioned observability risk adds value beyond full-state references, reduced-information PPO, heuristic risk, and state-only learned risk.

REFERENCES

- [1] Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. Constrained policy optimization. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 22–31. PMLR, 2017.
- [2] Moulay A. Akhloufi, Sebastian Arola, and Alexandre Bonnet. Drones chasing drones: Reinforcement learning and deep search area proposal. *Drones*, 3(3):58, 2019.
- [3] Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 2018.
- [4] Andrea Baisero, Brett Daley, and Christopher Amato. Asymmetric DQN for partially observable reinforcement learning. In *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence*, volume 180 of *Proceedings of Machine Learning Research*, pages 107–117. PMLR, 2022.
- [5] Jiayu Chen, Chao Yu, Guosheng Li, Wenhao Tang, Shilong Ji, Xinyi Yang, Botian Xu, Huazhong Yang, and Yu Wang. Online planning for multi-UAV pursuit-evasion in unknown environments using deep reinforcement learning. *IEEE Robotics and Automation Letters*, 10(8):8196–8203, 2025.
- [6] Jiaming Cheng, Ni Li, Changyin Dong, and Chong Tang. Hierarchical mental-states reasoning with dynamic fusion world models for imperfect multi-unmanned aerial vehicle cooperative-competitive environments. *Engineering Applications of Artificial Intelligence*, 173:114402, 2026.
- [7] Daniel Ebi, Gaspard Lambrechts, Damien Ernst, and Klemens Böhm. Informed asymmetric actor-critic: Leveraging privileged signals beyond full-state access, 2026.
- [8] Francisco Giral, Ignacio Gomez, and Soledad Le Clainche. Intercepting unauthorized aerial robots in controlled airspace using reinforcement learning. *Aerospace Science and Technology*, 168:110985, 2026.
- [9] Ju-Hyun Kim and Seungki Min. Risk-sensitive policy optimization via predictive CVaR policy gradient, 2024.
- [10] Jianan Li, Zhikun Wang, Susheng Ding, Shiliang Guo, and Shiyu Zhao. Cooperative bearing-only target pursuit via multiagent reinforcement learning: Design and experiment, 2025.
- [11] Yunxiao Lian, Ni Li, Pan Zhou, Ruiguang Hu, and Changyin Dong. A curriculum-guided and explainable reinforcement learning framework for fixed-wing unmanned aerial vehicle autopilots. *Engineering Applications of Artificial Intelligence*, 163:113015, 2026.
- [12] Xiong Liqin, Chen Xiliang, Luo Xijian, and Cao Lei. A perception-enhanced multi-agent deep reinforcement learning method for multi-UAV cooperative pursuit. *Expert Systems with Applications*, 311:131334, 2026.
- [13] Yushen Liu, Yin-Jen Chen, Ziyi Chen, Tao Wang, Heng Huang, Xugui Zhou, and Yanfu Zhang. Action-conditioned risk gating for safety-critical control under partial observability, 2026.
- [14] Delin Luo, Zihao Fan, Ziyi Yang, and Yang Xu. Multi-UAV cooperative maneuver decision-making for pursuit-evasion using improved MADRL. *Defence Technology*, 35:187–197, 2024.
- [15] Xiaoteng Ma, Junyao Chen, Li Xia, Jun Yang, Qianchuan Zhao, and Zhengyuan Zhou. DSAC: Distributional soft actor-critic for risk-sensitive reinforcement learning. *Journal of Artificial Intelligence Research*, 83, 2025.
- [16] Yecheng Jason Ma, Andrew Shen, Osbert Bastani, and Dinesh Jayaraman. Conservative and adaptive penalty for model-based safe reinforcement learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(5):5404–5412, 2022.
- [17] Pawel Miera, Hubert Szolc, and Tomasz Kryjak. LiDAR-based drone navigation with reinforcement learning, 2023.
- [18] Zhe Peng, Guohua Wu, Biao Luo, and Ling Wang. Multi-UAV cooperative pursuit strategy with limited visual field in urban airspace: A multi-agent reinforcement learning approach. *IEEE/CAA Journal of Automatica Sinica*, 12(7):1350–1367, 2025.
- [19] Lerrel Pinto, Marcin Andrychowicz, Peter Welinder, Wojciech Zaremba, and Pieter Abbeel. Asymmetric actor critic for image-based robot learning, 2017.
- [20] Alex Ray, Joshua Achiam, and Dario Amodei. Benchmarking safe exploration in deep reinforcement learning, 2019.
- [21] Leliang Ren, Yong Xian, Zhenyu Liu, Shaopeng Li, Daqiao Zhang, and Weilin Guo. Maneuvering target interception via deep reinforcement learning guidance using only line-of-sight rate measurement. *Engineering Applications of Artificial Intelligence*, 149:110523, 2025.
- [22] Alejandro Sanchez Roncero, Yixi Cai, Olov Andersson, and Petter Ögren. Learned controllers for agile quadrotors in pursuit-evasion games, 2025.
- [23] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017.
- [24] Kunal Shah and Mac Schwager. GRAPE: Geometric risk-aware pursuit-evasion. *Robotics and Autonomous Systems*, 121:103246, 2019.
- [25] Adam Stooke, Joshua Achiam, and Pieter Abbeel. Responsive safety in reinforcement learning by PID lagrangian methods, 2020.
- [26] Mulai Tan, Haocheng Sun, Dali Ding, Huan Zhou, and Yongli Liu. Scalable pursuit-evasion game for multi-fixed-wing UAV based on dynamic target assignment and hierarchical reinforcement learning. *Drones*, 10(1):5, 2025.
- [27] Brijen Thananjeyan, Ashwin Balakrishna, Suraj Nair, Michael Luo, Krishnan Srinivasan, Minh Hwang, Joseph E. Gonzalez, Julian Ibarz, Chelsea Finn, and Ken Goldberg. Recovery RL: Safe reinforcement learning with learned recovery zones. *IEEE Robotics and Automation Letters*, 6(3):4915–4922, 2021.
- [28] Yizhuo Wang, Yuhong Cao, Jimmy Chiun, Subhadeep Koley, Mandy Pham, and Guillaume Sartoretti. ViPER: Visibility-based pursuit-evasion via reinforcement learning. In *Proceedings of the 8th Conference on Robot Learning*, 2025.
- [29] Yuming Xiang, Sizhao Li, Rongpeng Li, Zhifeng Zhao, and Honggang Zhang. Decentralized consensus inference-based hierarchical reinforcement learning for multiconstrained UAV pursuit-evasion game. *IEEE Transactions on Neural Networks and Learning Systems*, 36(10):18229–18243, 2025.
- [30] Jiaping Xiao and Mir Feroskhan. Learning multipursuit evasion for safe targeted navigation of drones. *IEEE Transactions on Artificial Intelligence*, 5(12):6210–6224, 2024.
- [31] Tian Yan, Can Liu, Mengjing Gao, Zijian Jiang, and Tong Li. A deep reinforcement learning-based intelligent maneuvering strategy for the high-speed UAV pursuit-evasion game. *Drones*, 8(7):309, 2024.
- [32] Kun Yang, Ao Shen, Nengwei Xu, Fang Deng, Maobin Lu, and Chen Chen. A review of reinforcement learning approaches for pursuit-evasion games. *Chinese Journal of Aeronautics*, 39(6):103940, 2026.
- [33] Jiarui Zhang, Chengyong Lei, Chengjiang Dai, Lijie Wang, Zhichao Han, and Fei Gao. High-speed vision-based flight in clutter with safety-shielded reinforcement learning, 2026.
- [34] Ruilong Zhang, Qun Zong, Xiuyun Zhang, Liqian Dou, and Bailing Tian. Game of drones: Multi-UAV pursuit-evasion game with online motion planning by deep reinforcement learning. *IEEE Transactions on Neural Networks and Learning Systems*, 34(10):7900–7909, 2023.
- [35] Wei Zhang, Ke Song, Xuewen Rong, and Yibin Li. Coarse-to-fine UAV target tracking with deep reinforcement learning. *IEEE Transactions on Automation Science and Engineering*, 16(4):1522–1530, 2019.

- [36] Yang Zhao, Zidong Nie, Kangsheng Dong, Qinghua Huang, and Xuelong Li. Autonomous decision making for UAV cooperative pursuit-evasion game with reinforcement learning, 2024.
- [37] Yang Zhao, Wenzhe Zhao, and Xuelong Li. MSMAR-RL: Multi-step masked-attention recovery reinforcement learning for safe maneuver decision in high-speed pursuit-evasion game. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pages 311–319, 2025.
- [38] Canlun Zheng, Yize Mi, Hanqing Guo, Huaben Chen, and Shiyu Zhao. Vision-based cooperative MAV-capturing-MAV, 2025.
- [39] Minnan Zhou, Mustafa Shaikh, Vatsalya Chaubey, Patrick Haggerty, Shumon Koga, Dimitra Panagou, and Nikolay Atanasov. Control strategies for pursuit-evasion under occlusion using visibility and safety barrier functions. In *2025 IEEE International Conference on Robotics and Automation*, pages 12863–12869, 2025.